LESSON PLAN

GAS TURBINE ENGINE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	15
Total No. of Lesson Plans	16	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Ice protection, hot air protection, electrical system
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Ice protection, hot air protection, electrical system
5. Specific Objectives	13.1.7.1 Trainee will be able to explain various methods of Ice protection system.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of engine electrical schematics and ice protection logic diagrams; Safety: Ensure no live electrical components are touched; Instructor supervision: Monitor trainee understanding of system interconnections.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary hazards of ice accumulation on engine components?
(b) How does hot air protection prevent ice formation on the inlet?
(c) Which electrical systems are critical for monitoring engine performance?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Ice formation mechanisms on engine surfaces • Hot air bleed system operation • Electrical power distribution for protection systems	L & Ds	PPT, IB	15 min
(b)	• Anti-icing vs de-icing system differences • Thermal management of hot air valves • Electrical fault detection in protection circuits	L & Ds	PPT, PDF	20 min
(c)	• System integration of ice protection controls • Maintenance checks for electrical sensors • Operational limits for hot air systems	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the sequence of activation for the hot air protection system during icing conditions.
(b) Describe the role of electrical sensors in monitoring ice protection system status.
(c) Identify two maintenance checks required for the electrical system of the ice protection unit.

12. Home Assignment and Follow Up	Prepare a brief summary of the engine thrust calculation parameters to be discussed in the next lesson plan (LP 16).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Calculation of engine thrust, various parameters and tests, adjustments of engine controls	Refer to the master syllabus reference list	LP 16

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min